

Verification of whether the final matrix is in rref

1. Row 1 and 2 are non-zero rows. Row 3 is a zero row. $\therefore$ The condition that all non-zero rows must be above the zero rows is True.

2. Row 1 Leading Element = 1 in Column 1

Row 2 Leading Element = 1 in Column 2

$\therefore$ The condition that for each row, the leading element must be in a column to the right of the columns of the leading elements of all rows above it is True.

3. Row 2 Column 1 = Row 3 Column 1 = 0

Row 3 Column 2 = 0

$\therefore$ The condition that for each column having a leading entry all positions below the leading entry position must contain 0 is True.

4. The condition that each leading entry must be 1 is True.

5. Row 1 Column 2 = 0

$\therefore$ The condition that for each column having a leading entry all positions above the leading entry position must contain 0 is True.

$\therefore$ The matrix $\begin{bmatrix} 1 & 0 & -2 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ is in rref. $\square$

Column 3 has no pivots. $\therefore$ By defn, x_3 is a free variable. $\therefore$ By Theorem 2, the corresponding system of equations has infinitely many solutions.

$\therefore$ The vectors v_1, v_2 and v_3 are not linearly independent. $\square$

b) The corresponding system of equations to the final matrix in rref:

$$\textcircled{1} x_1 - 2x_3 = 0 \quad \textcircled{2} x_2 + x_3 = 0$$

The solution set is $x_1 = 2x_3, x_2 = -x_3$ and x_3 is free.

Let $x_3 = 1$. Then $x_2 = -1, x_1 = 2$. Substituting these values in the

original linear independence relation, we get: $\boxed{2v_1 - v_2 + v_3 = 0}$.
This is one (out of infinitely many) possible linear dependence relations among v_1, v_2 and v_3 . $\square$